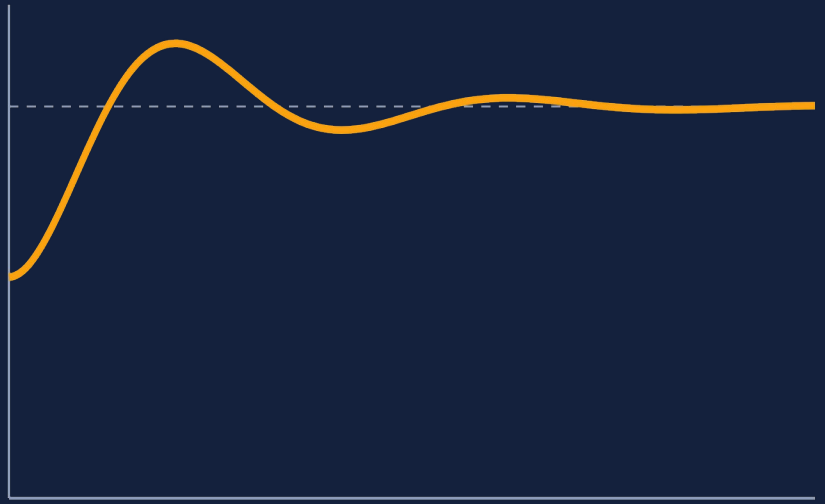
P, PI and PID Controllers



Control Systems (22EEG53)

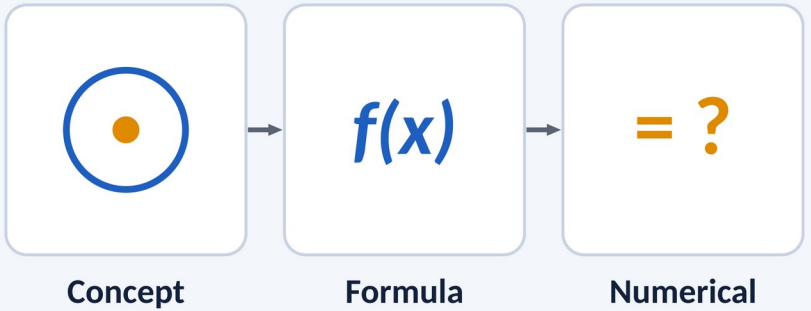
Unit 5: Compensators and Controllers

Topic: Proportional, integral and derivative control

Prof. Mitesh Astik

Learning Outcomes

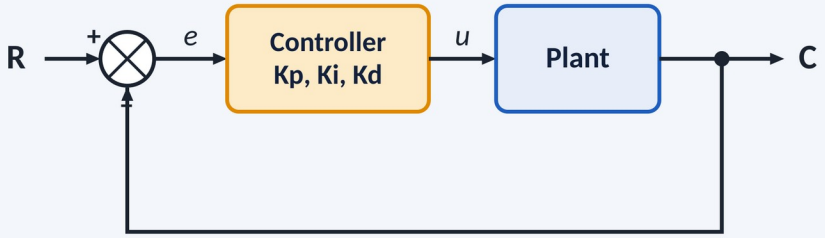
- Write P, PI, PD and PID control laws.
- Explain what each term does to the response.
- Tune a PID controller by Ziegler–Nichols.
- Compute Kp, Ki and Kd for a given plant test.



Linked to CO5
Design the appropriate compensator and controller for given specifications.

Recap: What You Already Know

- Error e = r − c drives the controller.
- Steady-state error depends on system type.
- Integrator adds a pole at the origin.
- Derivative responds to rate of change.



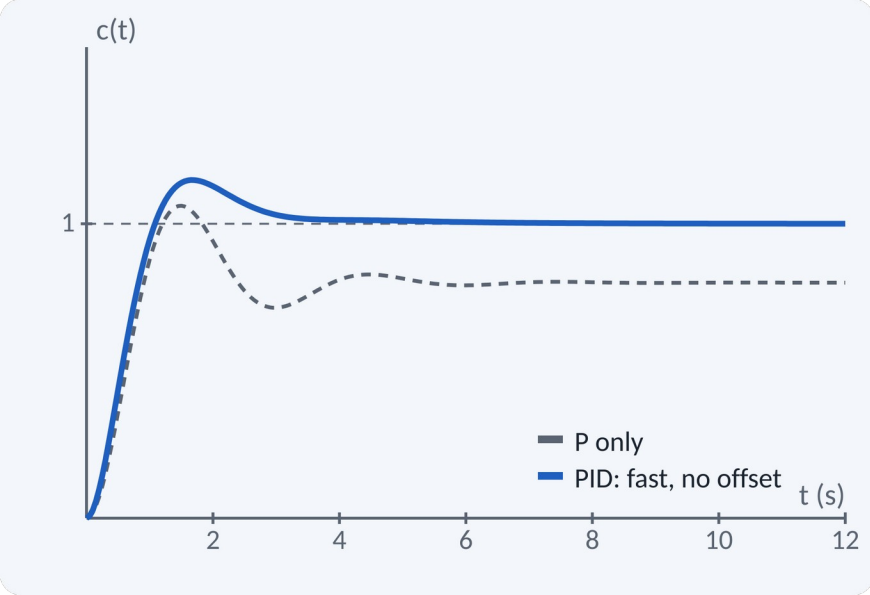
Error e = r − c → controller → control signal u

Most industrial loops use PID

Laplace: ∫e dt → E/s; de/dt → sE; Type N = number of integrators

Motivation: The Workhorse of Industry

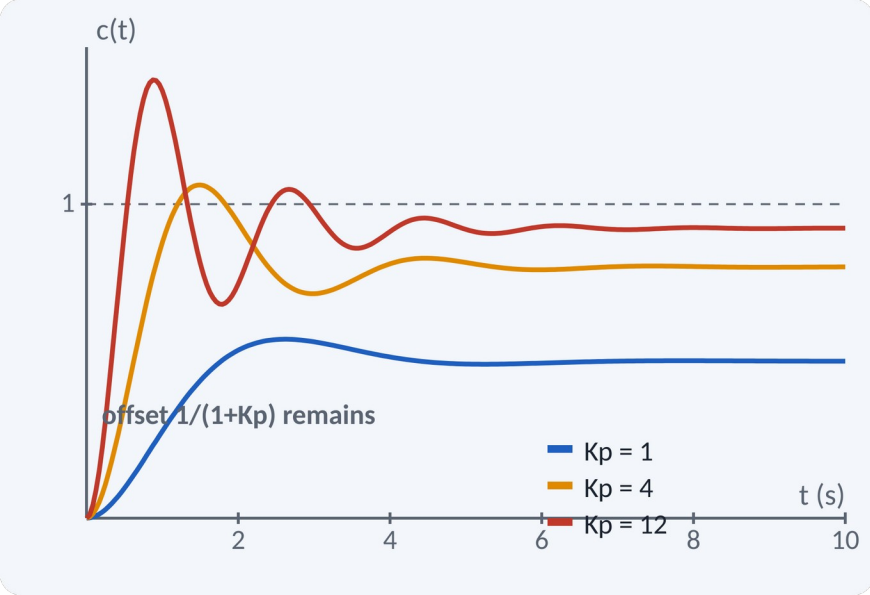
- Over 90 % of industrial loops use PID.
- Temperature, speed, level, pressure control.
- Three tuning knobs: Kp, Ki, Kd.
- Lab: PID design and DC position control.



PID gives fast response and zero steady-state error with little overshoot

Proportional (P) Controller

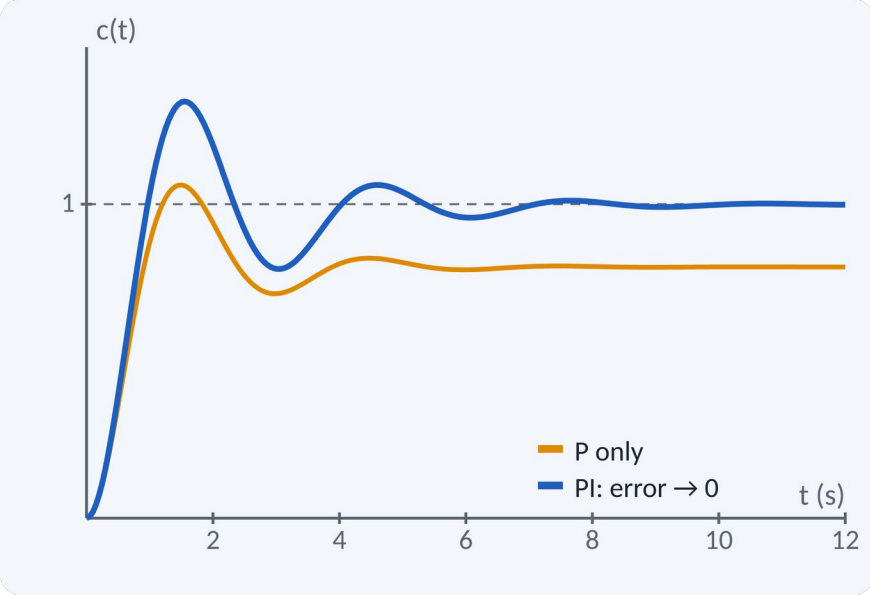
- Output proportional to the error.
- Raising Kp speeds up the response.
- Too large Kp gives overshoot and oscillation.
- A steady-state offset remains for Type 0 plants.



u(t) = Kp·e(t); Gc(s) = Kp
ess(step) = 1/(1 + Kp·G(0)) for a Type 0 plant

Proportional–Integral (PI) Controller

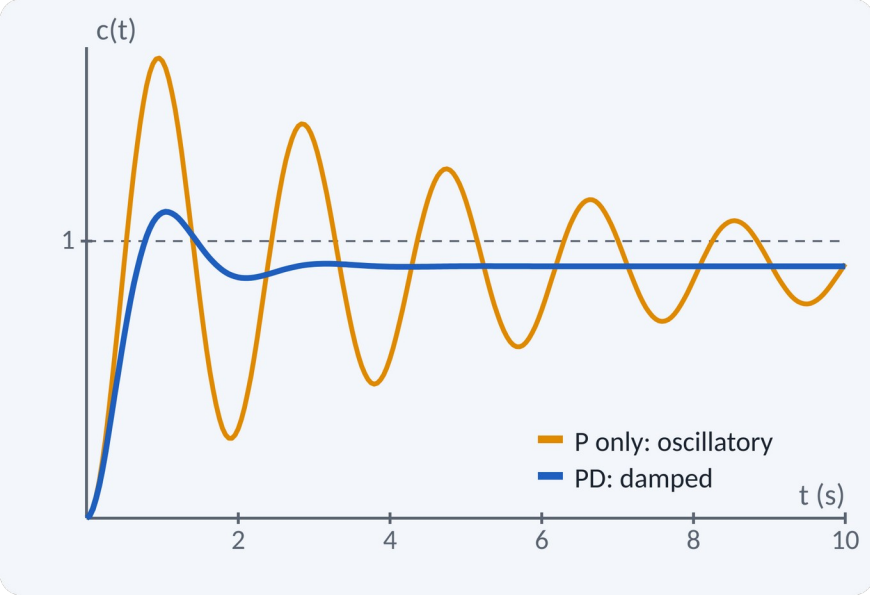
- Integral term accumulates the error.
- Output keeps changing until the error is zero.
- Eliminates the steady-state error.
- Adds lag: may increase overshoot and slow down.
- Pole at origin raises system type by one.



$$u = K_p \cdot e + K_i \int e\,dt; \quad G_c(s) = K_p + K_i/s = K_p(1 + 1/(T_i s)), \quad T_i = K_p/K_i$$

Proportional–Derivative (PD) Controller

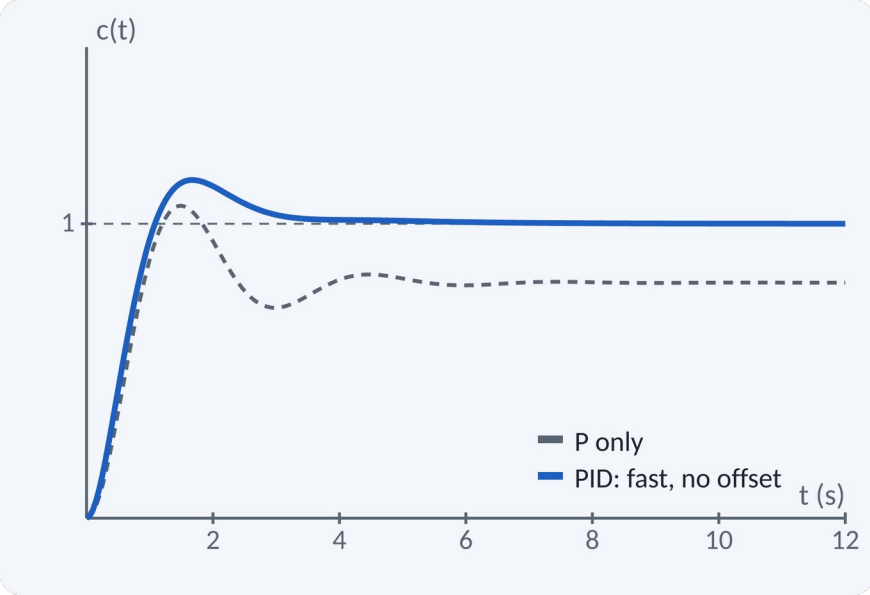
- Derivative term reacts to the rate of change of error.
- Adds damping: reduces overshoot.
- Acts like a lead compensator.
- Amplifies high-frequency noise.
- No effect on steady-state error.



u = Kp·e + Kd·de/dt; Gc(s) = Kp + Kd·s = Kp(1 + Td s), Td = Kd/Kp

PID Controller

- Combines proportional, integral and derivative actions.
- P gives speed; I removes offset; D adds damping.
- Standard form uses Ti and Td.
- Equivalent to a lag–lead compensator.



u = Kp[e + (1/Ti)∫e dt + Td·de/dt]; Gc(s) = Kp + Ki/s + Kd·s
Ki = Kp/Ti, Kd = Kp·Td

Effect of Each Gain

- Each term changes the response differently.
- Tune one gain at a time, small steps.
- Large Ki causes oscillation.
- Large Kd amplifies noise.

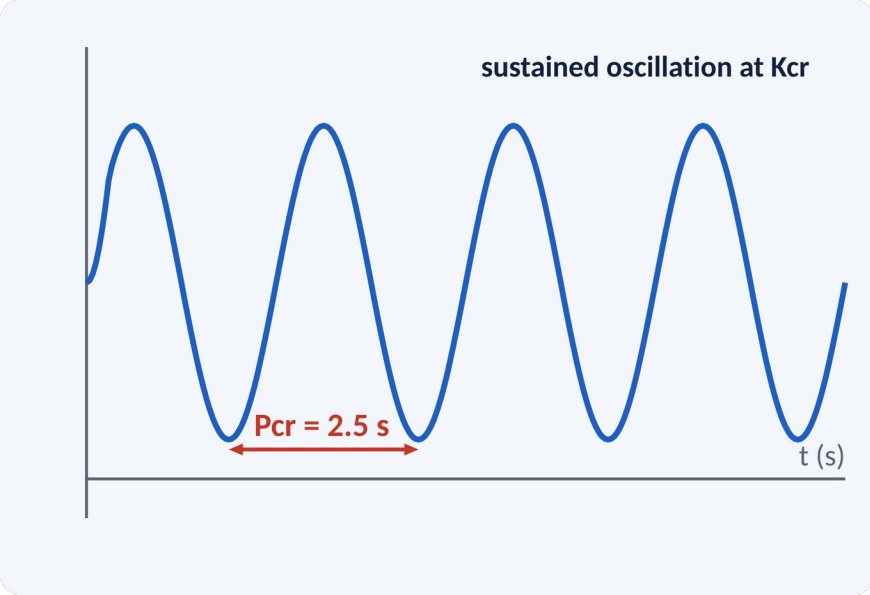
	Kp ↑	Ki ↑	Kd ↑
Rise time	decrease	decrease	small change
Overshoot	increase	increase	decrease
Settling time	small change	increase	decrease
Steady-state error	decrease	eliminate	no change

(effects are general trends; Kd amplifies noise)

Kp ↓ error, ↑ speed; Ki → zero error; Kd ↑ damping

Ziegler–Nichols Tuning

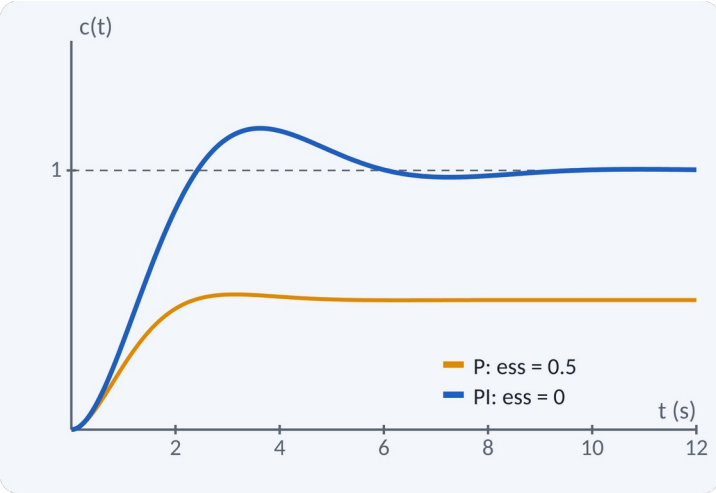
- Set Ki = Kd = 0; raise Kp until steady oscillation.
- Record the critical gain Kcr and period Pcr.
- Use the table to set Kp, Ti and Td.
- Gives a quick, reasonable first tuning.
- Fine-tune afterwards.



P: Kp = 0.5Kcr; PI: Kp = 0.45Kcr, Ti = Pcr/1.2
PID: Kp = 0.6Kcr, Ti = 0.5Pcr, Td = 0.125Pcr

Solved Example 1: P vs PI

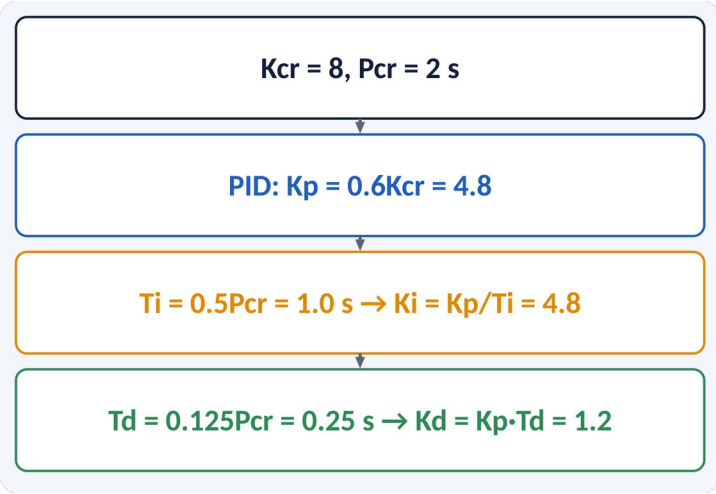
- Plant $G(s) = 1/(s^2 + 2s + 1)$, unity feedback. Compare steady-state error for a unit step with P (Kp = 1) and PI (Kp = 1, Ki = 1).
- 1. P: $ess = 1/(1 + Kp \cdot G(0)) = 1/(1 + 1) = 0.5$.
- 2. PI: $Gc \cdot G$ has a pole at the origin → Type 1.
- 3. $Kp(PI) = \infty$ so ess = 0.
- 4. Check stability: $s^3 + 2s^2 + 2s + 1 = 0$ → Routh: $2 \cdot 2 > 1$ → stable.



Answer: P gives ess = 0.5; PI gives ess = 0 (stable)

Solved Example 2: Ziegler–Nichols PID

- A P-only test gives sustained oscillation at Kcr = 8 with period Pcr = 2 s. Tune a PID.
- 1. Kp = 0.6Kcr = 4.8.
- 2. Ti = 0.5Pcr = 1.0 s → Ki = Kp/Ti = 4.8 s^{-1}.
- 3. Td = 0.125Pcr = 0.25 s → Kd = Kp·Td = 1.2 s.
- 4. Gc(s) = 4.8 + 4.8/s + 1.2s = 1.2(s^2 + 4s + 4)/s = 1.2(s + 2)²/s.



Answer: Kp = 4.8, Ki = 4.8, Kd = 1.2; Gc(s) = 1.2(s + 2)²/s

Common Mistakes and Exam Tips

Common mistakes

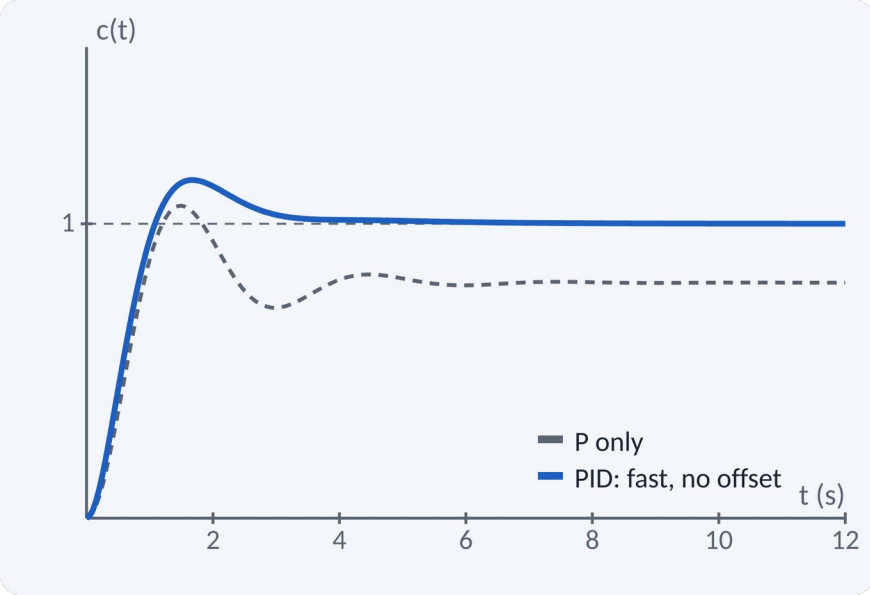
- Using the derivative term on a noisy signal without a filter.
- Expecting P control to remove steady-state error.
- Increasing Ki too much (oscillation).
- Mixing Ti with Ki (Ki = Kp/Ti).
- Applying Ziegler–Nichols to an unstable plant.

Exam tips

- Start with P, then add I, then D.
- Tune in small steps, one gain at a time.
- Remember: I removes offset, D adds damping.
- Check stability after every change.

Summary

- P: u = Kp·e — fast, leaves an offset.
- PI: removes steady-state error, adds lag.
- PD: adds damping, noisy.
- PID: Gc = Kp + Ki/s + Kd·s combines all.
- Z–N: Kp = 0.6Kcr, Ti = 0.5Pcr, Td = 0.125Pcr.



Gc(s) = Kp + Ki/s + Kd s = Kp(1 + 1/(Ti s) + Td s)

Quick Quiz

Q1. Which action removes steady-state error to a step?
(a) P (b) D (c) I (d) none

Ans: (c)

Q2. Derivative control mainly improves:
(a) accuracy (b) damping (c) type (d) gain

Ans: (b)

Q3. Ki in terms of Kp and Ti:
(a) Kp·Ti (b) Kp/Ti (c) Ti/Kp (d) Kp + Ti

Ans: (b)

Q4. Z–N PID gain Kp equals:
(a) 0.5Kcr (b) 0.45Kcr (c) 0.6Kcr (d) Kcr

Ans: (c)

Q5. A PI controller makes a Type 0 system:
(a) Type 0 (b) Type 1 (c) Type 2 (d) unstable always

Ans: (b)

References

1. **Benjamin C. Kuo, Automatic Control Systems**
   7th Ed., PHI, 2007 (Units 3, 4, 5 textbook)

2. **K. Ogata, Modern Control Engineering**
   5th Ed., Pearson Education Asia / PHI, 2009 (reference)

3. **J. J. D'Azzo and C. H. Houpis, Feedback Control System Analysis and Synthesis**
   4th Ed., McGraw Hill International, 1999 (reference, Unit 5)

Next lecture: state-variable analysis of physical systems